

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455630.9089 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.0068
Transit depth (Rp/Rs)^2	2.76 +/- 0.23 [%]
Orbital Inclination (inc)	86.35 +/- 1.15
Airmass coefficient 1 (a1)	1.367 +/- 0.011
Airmass coefficient 2 (a2)	-0.252 +/- 0.048
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	None
Optimal Aperture	4.18
Optimal Annulus	12.58
Transit Duration (day)	0.093 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.166 $\pm$ 0.0068 vs 0.1457 (deviation 2.62 σ)
Duration fit vs published	0.093 d vs 0.12 d (deviation 0.99 σ)
Mid-time O–C	20.8 min
Depth SNR	21.5
Residual scatter	0.8 %

Light curve

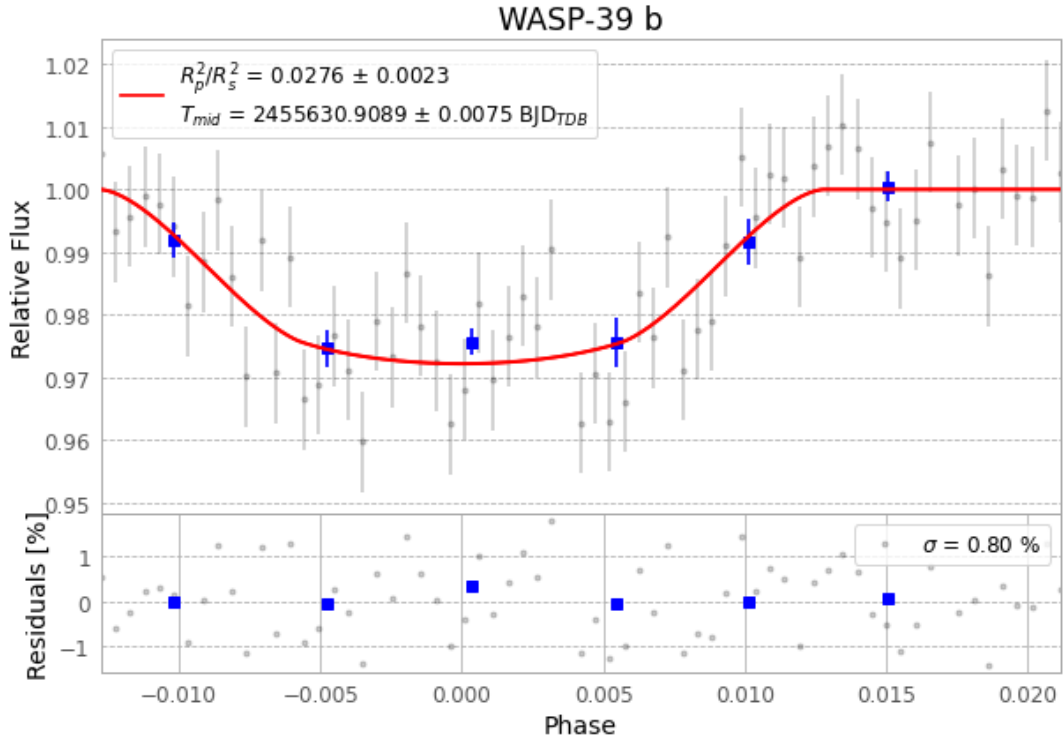


Figure 1 — FinalLightCurve_WASP-39 b_2011-03-10.png (SHA-256 1d99edf61e877ec0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [139, 138], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 22316e5a237a5323...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), WASP-39110310082713.FITS → WASP-39110310114513.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-110310.log (055955359a7b...), FinalLightCurve_WASP-39 b_2011-03-10.png (1d99edf61e87...), FinalLightCurve_WASP-39 b_2011-03-10.csv (bf7930ec582b...)

Compute resources

Wall time 92.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 00:26Z.